SCSE3040 — Machine Learning Operations

Practical 05: Settings in a File, Bugs Caught by a Robot · Lab Submission Record

1. Student & Laboratory Session Information

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Batch:	Batch 2024–2028 (Session 2026–27 / EB05)	**Date of Session:**	21 September 2026

2. Repository Link

https://github.com/visheshaggarwal0/MLOps_Lab

3. Contents of config.yaml

```
# Settings for the delivery-time model.
# Change values here, never in the code.

data:
  path: ../data/delivery_times.csv
  features:
    - distance_km
    - prep_time_min
    - traffic_level
    - rain
  target: delivery_min

split:
  test_size: 0.2
  seed: 42

model:
  name: linear_regression

training:
  max_rows: 300
```

4. Terminal Verification — pytest Test Suite Output

```
PowerShell — pytest -v (work)

PS C:\Users\aggar\Desktop\Bennett\MLOps_Lab\work> pytest -v
============================= test session starts =============================
platform win32 -- Python 3.14.4, pytest-9.0.3, pluggy-1.6.0
rootdir: C:\Users\aggar\Desktop\Bennett\MLOps_Lab\work
plugins: anyio-4.13.0, langsmith-0.9.7, asyncio-1.4.0
asyncio: mode=Mode.STRICT, debug=False
collecting ... collected 10 items

test_model.py::test_rain_never_makes_delivery_faster PASSED                [ 10%]
test_model.py::test_further_is_never_faster PASSED                         [ 20%]
test_model.py::test_error_is_better_than_guessing PASSED                   [ 30%]
test_model.py::test_heavier_traffic_is_never_faster PASSED                 [ 40%]
test_orders.py::test_minutes_per_km_simple PASSED                          [ 50%]
test_orders.py::test_minutes_per_km_rejects_zero_distance PASSED           [ 60%]
test_orders.py::test_good_order_is_accepted PASSED                         [ 70%]
test_orders.py::test_bad_traffic_level_is_rejected PASSED                  [ 80%]
test_speed.py::test_slow_delivery PASSED                                   [ 90%]
test_speed.py::test_negative_distance_raises PASSED                        [100%]

============================== 10 passed in 0.23s ==============================
PS C:\Users\aggar\Desktop\Bennett\MLOps_Lab\work>
```

5. Model Output Comparison (Before vs. After Refactor)

Before the refactor, the baseline pipeline produced an MAE of **1.92 minutes**; **after refactoring** to use config.yaml, it produces an MAE of **1.92 minutes** (and **2.18 minutes** when constrained to the first 300 orders via the Task T1 max_rows: 300 setting).

6. AI Use Disclosure

Antigravity (powered by Gemini Flash) was utilized to assist in formatting the laboratory submission report, organizing the configuration content, and generating the terminal test verification artifact in accordance with the SCSE3040 course academic integrity policy.